

Recap:

- Horizon: N - Policy value functions under any policy $\pi \in \Pi^{\text{MR}}$:

$$V_{N:N}^{\pi} : \mathcal{S} \rightarrow \mathbb{R}, \quad V_{N:N}^{\pi}(s) := E^{\pi} [R_N | \{S_{N-1} = s\}]$$

$$= \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \pi_{N-1}(a|s) P(s'|s,a) r_N(s,a,s')$$

For any $n \in \{1, \dots, N-1\}$,

$$V_{n:N}^{\pi} : \mathcal{S} \rightarrow \mathbb{R}, \quad V_{n:N}^{\pi}(s) := E^{\pi} \left[\sum_{t=n}^N R_t \mid \{S_{n-1} = s\} \right]$$

$$= E^{\pi} [R_n + V_{n+1:N}^{\pi}(S_n) \mid \{S_{n-1} = s\}]$$

$$r_n(S_{n-1}, A_{n-1}, S_n) = \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \pi_{n-1}(a|s) P(s'|s,a) [r_n(s,a,s') + V_{n+1:N}^{\pi}(s')]$$

- Policy-value functions under any $\pi \in \Pi^{\text{HR}}$:

$$V_{N:N}^{\pi} : \mathcal{S} \rightarrow \mathbb{R}, \quad V_{N:N}^{\pi}(s) := E^{\pi} [R_N | \{S_{N-1} = s\}]$$

$$= \sum_{\substack{\underline{h} \in \mathcal{H}_{N-1}: \\ \underline{h} \text{ ends in } s}} \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P^{\pi}(H_{N-1} = \underline{h}) \pi_{N-1}(a|\underline{h}) P(s'|s,a) r_N(s,a,s')$$

$$\underline{h} = (\dots, s)$$

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$$= E^{\pi} [R_n + V_{n+1:N}^{\pi}(S_n) \mid \{S_{n-1} = s\}]$$

$$= \sum_{\substack{\underline{h} \in \mathcal{H}_{n-1}: \\ \underline{h} \text{ ends in } s}} \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P^{\pi}(H_{n-1} = \underline{h}) \pi_{n-1}(a|\underline{h}) P(s'|s,a) [r_n(s,a,s') + V_{n+1:N}^{\pi}(s')]$$

- Optimal value function and Optimal Policy

$$\forall s \in \mathcal{S}, \quad V_{1:N}^*(s) = \sup_{\pi \in \Pi^{\text{HR}}} V_{1:N}^{\pi}(s), \quad V_{1:N}^{\pi^*}(s) = V_{1:N}^*(s) \quad \forall s \in \mathcal{S}$$

↙ Theorem 4.2, Chapter 4, Puterman and Chan

Theorem 1: Suppose that $\mathcal{A}$ is finite. Then, there exists $\pi^* \in \Pi^{\text{MD}}$ such that:

$$(1) \quad V_{1:N}^{\pi^*}(s) = \sup_{\pi \in \Pi^{\text{HR}}} V_{1:N}^{\pi}(s) \quad \forall s \in \mathcal{S}.$$

This shows that an optimal Markovian deterministic policy exists whenever $\mathcal{A}$ is finite.

Consequently, we may write

$$V_{1:N}^{\pi^*}(s) = \sup_{\pi \in \Pi^{\text{HR}}} V_{1:N}^{\pi}(s) = \max_{\pi \in \Pi^{\text{HR}}} V_{1:N}^{\pi}(s) = \max_{\pi \in \Pi^{\text{MD}}} V_{1:N}^{\pi}(s).$$

(2) The optimal policy π^* satisfies

$$V_{n:N}^{\pi^*}(s) = \sup_{\pi \in \Pi^{\text{HR}}} V_{n:N}^{\pi}(s) \quad \forall n \in \{1, \dots, N\}.$$

This shows that the optimal policy achieves the optimal value starting from any time epoch n .

[The proof of Theorem 1 will be covered later]

Computing the Optimal Value Function (Assuming Theorem 1)
and the Optimal Policy

- Define $V_{N:N}^* : \mathcal{S} \rightarrow \mathbb{R}$ by

$$V_{N:N}^*(s) = \sup_{\pi \in \Pi^{\text{MD}}} V_{N:N}^{\pi}(s) \quad \rightarrow \text{assuming Theorem 1 holds}$$

$$\pi_0(a|s) = \sup_{\pi \in \Pi^{\text{MD}}} \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \pi_{N-1}(a|s) P(s'|a, s) r_N(s, a, s')$$

$$\pi = \{\overset{\uparrow}{\pi_0}, \pi_1, \dots, \pi_{N-1}\} = \sup_{w \in \Delta(\mathcal{A})} \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} w(a) P(s'|s, a) r_N(s, a, s')$$

$$\downarrow$$

$$\pi_1(a|s)$$

$$= \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s'|s, a) r_N(s, a, s').$$

$$\pi_{N-1}^*(s) = \operatorname{argmax}_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s'|s, a) r_N(s, a, s')$$

- For any $n \in \{1, \dots, N-1\}$, using backwards induction, let

$$\begin{aligned}
 V_{n:N}^*(s) &= \sup_{\pi \in \Pi^{\text{MD}}} V_{n:N}^\pi(s) \\
 &\quad \text{assuming Theorem 1} \\
 &= \sup_{\pi \in \Pi^{\text{MD}}} E^\pi \left[R_n + V_{n+1:N}^\pi(s_n) \mid \{s_{n-1} = s\} \right] \\
 &\quad \text{assuming Theorem 1} \\
 &= \sup_{\pi \in \Pi^{\text{MD}}} \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \pi_{n-1}(a|s) P(s'|s, a) \left[r_n(s, a, s') + V_{n+1:N}^\pi(s') \right] \\
 &\quad \text{assuming Theorem 1} \\
 &= \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s'|s, a) \left[r_n(s, a, s') + V_{n+1:N}^*(s') \right]. \quad \text{assuming Theorem 1}
 \end{aligned}$$

$n = N-1$

also, let $\pi_{n-1}^*: \mathcal{S} \rightarrow \mathcal{A}$ be defined by

$$\pi_{n-1}^*(s) = \arg \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s'|s, a) \left[r_n(s, a, s') + V_{n+1:N}^*(s') \right].$$

Return Policy : $\pi^* = \{\pi_0^*, \dots, \pi_{N-1}^*\}$

Return Value function : $V_{1:N}^*$

Proposition 1 :

- ① The value functions returned by the above algorithm ($V_{1:N}^*, \dots, V_{N:N}^*$) are optimal in Π^{MD} for each time epoch n , i.e.,

$$V_{n:N}^*(s) = \sup_{\pi \in \Pi^{\text{MD}}} V_{n:N}^\pi(s) \quad \forall s \in \mathcal{S}.$$

- ② The policy π^* returned by the above algorithm is optimal in the class Π^{MD} .

Proof of Proposition 1 :

(i.e., $V_{n:N}^*(s) = V_{n:N}^{\pi^*}(s)$)

- ① We will prove by induction on n .

* The result holds trivially for $n=N$ because

$$V_{N:N}^*(s) = \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s'|s, a) r_N(s, a, s')$$

$$\geq \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \pi_{n-1}(a|s) \cdot P(s'|s, a) r_N(s, a, s') \quad \forall \pi \in \Pi^{\text{MD}}$$

* Assuming that $V_{n:N}^*(s) \geq V_{n:N}^\pi(s) \quad \forall s \in \mathcal{S}, \forall \pi \in \Pi^{\text{MD}}$, we have

For any $s \in \mathcal{S}$:

$$\begin{aligned} V_{n-1:N}^*(s) &= \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s'|s, a) [r_{n-1}(s, a, s') + V_{n:N}^*(s')] \\ &\geq \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s'|s, a) [r_{n-1}(s, a, s') + V_{n:N}^\pi(s')] \quad \forall \pi \in \Pi^{\text{MD}} \\ &\geq \underbrace{\sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \pi_{n-1}(a|s) P(s'|s, a) [r_{n-1}(s, a, s') + V_{n:N}^\pi(s')]}_{= V_{n-1:N}^\pi(s)} \quad \forall \pi \in \Pi^{\text{MD}} \end{aligned}$$

Taking sup over $\pi \in \Pi^{\text{MD}}$ on both sides, we get

$$V_{n-1:N}^*(s) \geq \sup_{\pi \in \Pi^{\text{MD}}} V_{n-1:N}^\pi(s) \quad \forall s \in \mathcal{S}$$

② The proof of part ② is straightforward from the definition of π_n^* for each n .

Search for an Optimal Policy in Π^{HR}

- Given $\pi \in \Pi^{\text{HR}}$, define $u_{N:N}^\pi : \mathcal{H}_{N-1} \rightarrow \mathbb{R}$ as

$$\begin{aligned} u_{N:N}^\pi(h_{N-1}) &:= E^\pi [R_N | H_{N-1} = h_{N-1}] \\ &= \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \pi_{N-1}(a|h_{N-1}) \cdot P(s'|s_{N-1}, a) r_N(s_{N-1}, a, s') \end{aligned}$$

(s₀, a₀, ..., s_{N-1})

- Given $\pi \in \Pi^{\text{HR}}$ and $n \in \{1, \dots, N-1\}$, define $u_{n:N}^\pi : \mathcal{H}_{n-1} \rightarrow \mathbb{R}$ as

$$\begin{aligned} u_{n:N}^\pi(h_{n-1}) &:= E^\pi \left[\sum_{t=n}^N R_t \mid H_{n-1} = h_{n-1} \right] \\ &= \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \pi_{n-1}(a|h_{n-1}) P(s'|s_{n-1}, a) [r_n(s_{n-1}, a, s') + u_{n+1:N}^\pi(h_{n-1}, a, s')] \end{aligned}$$

(s₀, a₀, ..., s_{n-1})

- Let $u_{n:N}^* : \mathcal{H}_{n-1} \rightarrow \mathbb{R}$ be defined as

$$u_{n:N}^*(h_{n-1}) = \sup_{\pi \in \Pi_{HR}} u_{n:N}^\pi(h_{n-1}),$$

- For each $n \in \{1, \dots, N-1\}$, let $u_{n:N}^* : \mathcal{H}_{n-1} \rightarrow \mathbb{R}$ be defined as

$$u_{n:N}^*(h_{n-1}) = \sup_{\pi \in \Pi_{HR}} u_{n:N}^\pi(h_{n-1}), \quad h_{n-1} \in \mathcal{H}_{n-1}.$$

Proposition 2: The functions $\{u_{n:N}^*\}_{n=1}^N$ satisfy the Bellman equations:

(1) $u_{n:N}^*(h_{n-1}) = \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s' | s_{n-1}, a) r_N(s_{n-1}, a, s')$

$(s_0, a_0, \dots, s_{N-1})$

Consequently, $u_{n:N}^*(h_{n-1})$ = a function of s_{n-1} only

(2) For all $n \in \{1, \dots, N-1\}$,

$$u_{n:N}^*(h_{n-1}) = \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s' | s_{n-1}, a) [r_n(s_{n-1}, a, s') + u_{n+1:N}^*(h_{n-1}, a, s')]$$

For each $n \in \{1, \dots, N-1\}$,

$$u_{n:N}^*(h_{n-1}) = \text{a function of } s_{n-1} \text{ only} \quad \forall h_{n-1} \in \mathcal{H}_{n-1}.$$

Proof of Proposition 2:

$$u_{n:N}^*(h_{n-1}) = \sup_{\pi \in \Pi_{HR}} \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} \pi_{n-1}(a | h_{n-1}) \cdot P(s' | s_{n-1}, a) r_N(s_{n-1}, a, s')$$

$(s_0, a_0, \dots, s_{N-1})$

$$= \sup_{w \in \Delta(\mathcal{A})} \sum_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} w(a) P(s' | s_{n-1}, a) r_N(s_{n-1}, a, s')$$

$$= \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} w(a) P(s' | s_{n-1}, a) r_N(s_{n-1}, a, s')$$

→ this shows that when $\mathcal{A}$ is finite, we can focus our attention on

deterministic policies only when trying to find the optimal policy.

The proof of part (2) is left as exercise.

↳ Hint : Induction !

Lastly, we have the following important result.

Theorem 2 : We have

$$u_{n:N}^* \left(\overset{(s_0, a_0, \dots, s_{n-1})}{h_{n-1}} \right) = v_{n:N}^* (s_{n-1}) \quad \forall h_{n-1} = (s_0, a_0, \dots, s_{n-1}) \in \mathcal{H}_{n-1}$$

In particular,

$$\forall n \in \{1, \dots, N\}.$$

$$u_{1:N}^*(s) = v_{1:N}^*(s) \quad \forall s \in \mathcal{S}.$$

Proof of Theorem 2 : Induction.

Base case $n=N$:

$$\begin{aligned} u_{N:N}^*(h_{N-1}) &= \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s' | s_{N-1}, a) r_N(s_{N-1}, a, s') \\ &= v_{N:N}^*(s_{N-1}). \end{aligned}$$

Assume that $u_{n:N}^*(h_{n-1}) = v_{n:N}^*(s_{n-1}) \quad \forall h_{n-1} = (s_0, a_0, \dots, s_{n-1}) \in \mathcal{H}_{n-1}.$

Then,

$$\begin{aligned} u_{n-1:N}^*(h_{n-2}) &= \max_a \sum_{s' \in \mathcal{S}} P(s' | s_{n-2}, a) \left[r_{n-1}(s_{n-2}, a, s') + u_{n:N}^*(h_{n-2}, a, s') \right] \\ &= \max_a \sum_{s' \in \mathcal{S}} P(s' | s_{n-2}, a) \left[r_{n-1}(s_{n-2}, a, s') + v_{n:N}^*(s') \right] \\ &= v_{n-1:N}^*(s_{n-2}) \end{aligned}$$

Theorem 2 and Proposition 2 together imply that

$$\sup_{\pi \in \Pi^{\text{HR}}} V_{1:N}^{\pi}(s) = \sup_{\pi \in \Pi^{\text{HD}}} V_{1:N}^{\pi}(s) \quad \forall s \in \mathcal{S}.$$

Put together with Theorem 1, we get

$$\sup_{\pi \in \Pi^{\text{HR}}} V_{1:N}^{\pi}(s) = \sup_{\pi \in \Pi^{\text{HD}}} V_{1:N}^{\pi}(s) = \sup_{\pi \in \Pi^{\text{MR}}} V_{1:N}^{\pi}(s) = \sup_{\pi \in \Pi^{\text{MD}}} V_{1:N}^{\pi}(s) \quad \forall s \in \mathcal{S}.$$
